



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to LAKSHMI PETROL BUNK

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

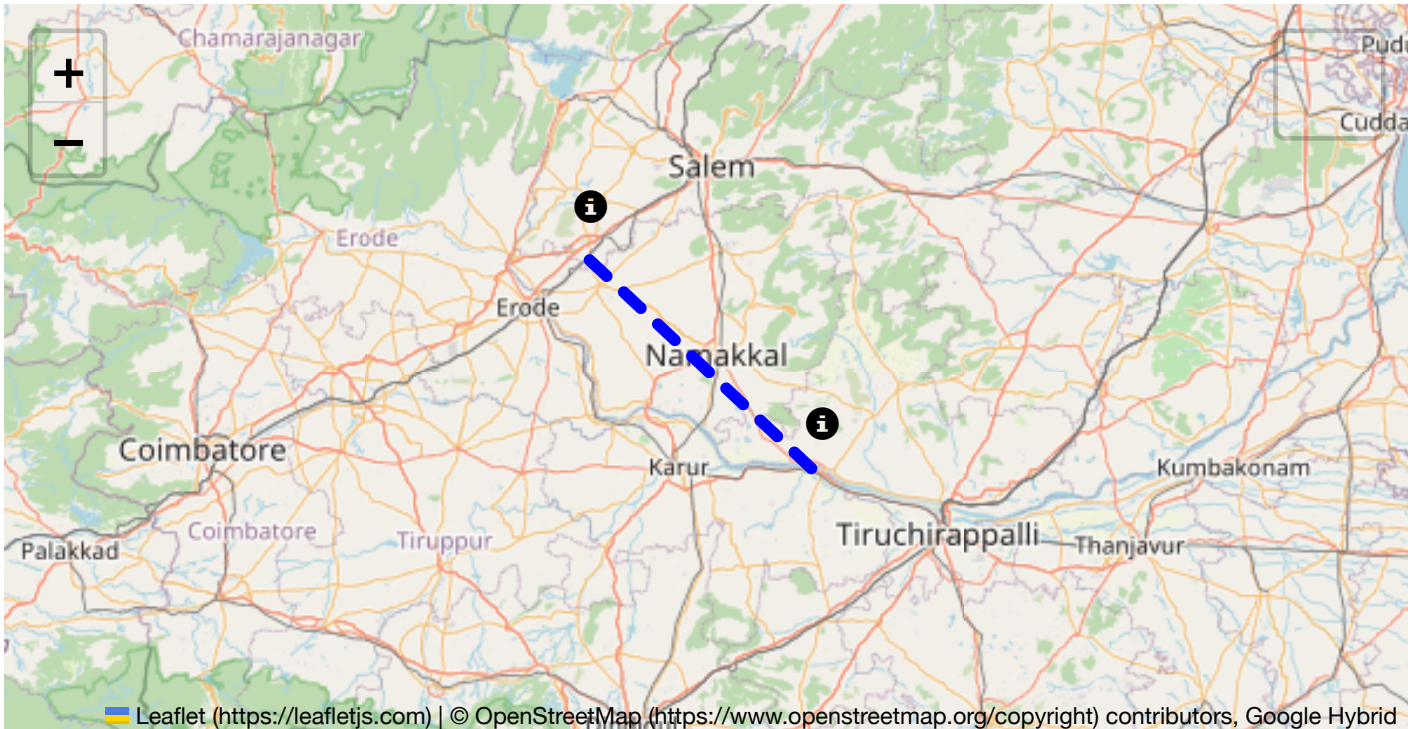
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

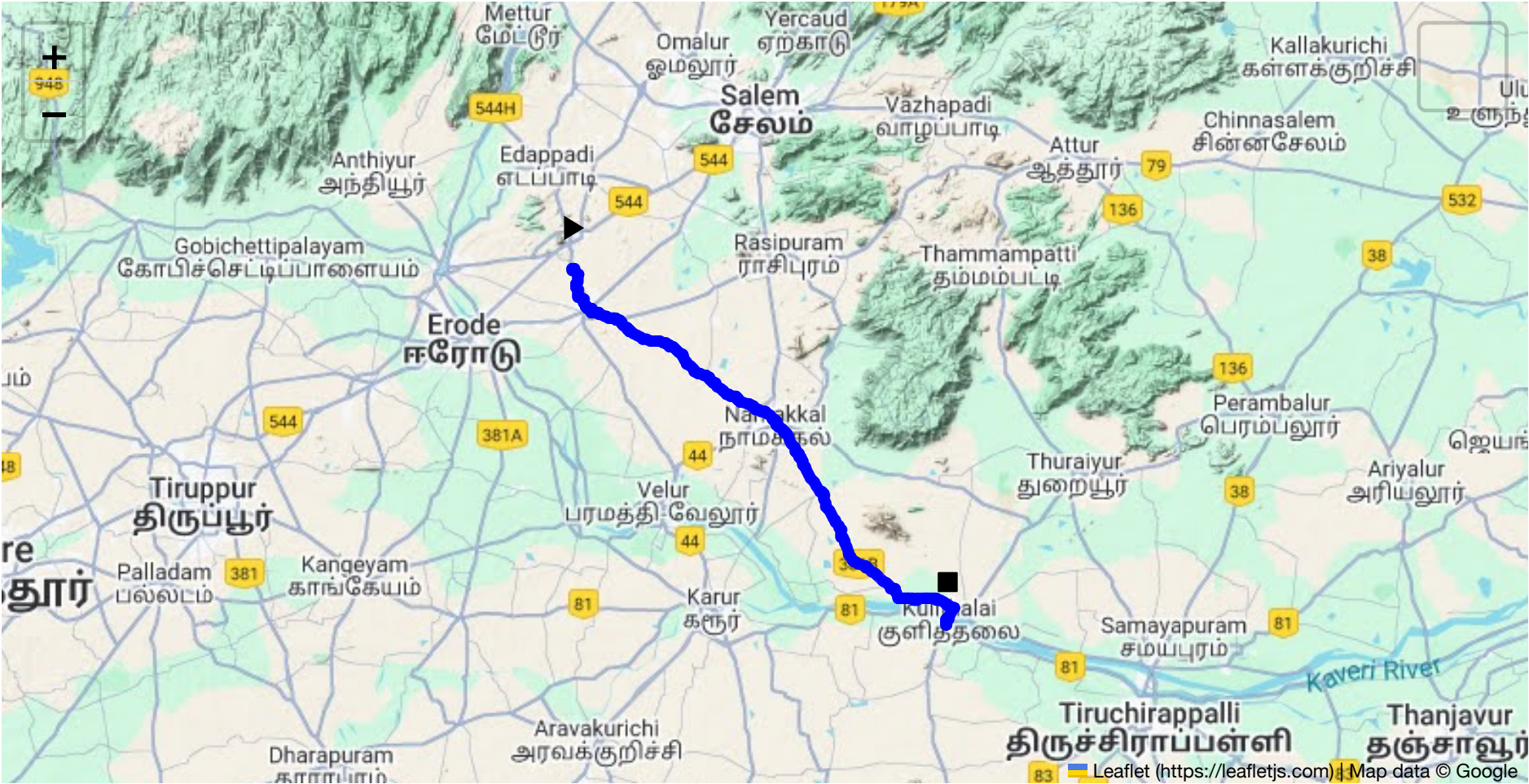
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 91.35 km
Estimated Duration: 1.9 hours
Adjusted Duration (Heavy Vehicle): 2.4 hours

Start: (11.4381, 77.8734)
End: (10.934953, 78.413536)



Welcome to the Journey Risk Management Study

Route Analysis Report

1. Overview of the Route Map

The route from Sangagiri (CVQF+23W) to Kulithalai (WCM7+XC2) spans approximately 91.35 kilometers primarily along major roads, including the NH44 and SH71. The route traverses through both urban and rural areas, with varying traffic conditions and infrastructure quality. It is essential to pay attention to route-specific signage and GPS navigation for the most accurate directions.

2. Weather Conditions and Potential Hazards

The region experiences a tropical climate with hot summers, heavy monsoons, and mild winters.

- **Summer (March to June):** Temperatures can soar, leading to potential overheating of engines.
- **Monsoon (July to September):** Heavy rains may lead to waterlogging and slippery roads, increasing the risk of hydroplaning.
- **Winter (November to February):** Generally mild with less severe weather; however, occasional fog can reduce visibility.

3. Traffic Patterns and Congestion Areas

- **Peak Hours:** Traffic tends to peak during morning (7 AM - 10 AM) and evening (5 PM - 8 PM) hours.
- **Congestion-Prone Areas:** Kallakurchi and Karur tend to be congested. Alternative routes or departure times can aid in avoiding delays.

4. Road Quality and Infrastructure

Road quality is mixed along the route:

- **NH44:** Generally well-maintained with periodic rest stops and fueling stations.
- **SH71:** Some sections may have potholes and narrower lanes, particularly in rural stretches. Regular maintenance of vehicle suspension and tires is recommended to cope with variable road conditions.

5. Alternative Routes for Emergencies

- **Via NH79 and NH536:** An alternative if NH44 is congested or under maintenance. Slightly longer but useful in emergencies.
- **Local Detours:** Useful in bypassing minor urban traffic via smaller roads, though these may not be suitable for heavy trucks.

6. Local Regulations

Transporting hazardous materials requires compliance with state and central regulations, including:

- Permits for hazardous materials.
- Adherence to speed limits and transporter inspections at designated checkpoints.

7. Historical Incidents

The region has minimal recorded incidents involving hazardous materials. However, caution is necessary near densely populated urban centers and during severe weather conditions.

8. Environmental and Sensitive Areas

- **Reserved Forests and Water Bodies:** Ensure no leaks or spills to protect local flora, fauna, and water sources.
- **Agricultural Areas:** Near NH44 and SH71, minimize air and noise pollution in these sensitive areas.

9. Communication Coverage

Most segments of the route have reliable mobile network coverage, except:

- **Rural Stretches:** Potential dead zones in remote areas; carrying physical maps and ensuring vehicle communication systems are functioning can mitigate risk.

10. Emergency Response Times

- **Urban Areas (e.g., Kallakurchi, Karur):** 20-30 minutes for emergency response.
- **Rural Stretches:** Response times can extend to 45 minutes or more. Plan accordingly and carry an emergency kit.

12. Overall Summary of Risk Assessment

The route from Sangagiri to Kulithalai presents moderate risk primarily due to weather conditions, mixed road quality, and traffic congestion in urban areas. Preparation for weather variations, adherence to regulations, diligent vehicle maintenance, and strategic planning of travel times are crucial for minimizing hazards. Regular communication with emergency services and monitoring weather advisories will further enhance safety protocols.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.94342, 78.41370	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44019, 77.87547	15 KM/Hr
5	Turn	Medium	11.39685, 77.88861	30 KM/Hr
6	Turn	Medium	11.38342, 77.89975	30 KM/Hr
7	Turn	Medium	11.38295, 77.90001	30 KM/Hr
8	Turn	High	11.38186, 77.90258	15 KM/Hr
9	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
10	Turn	Medium	11.22411, 78.16637	30 KM/Hr
11	Turn	Medium	11.12316, 78.23249	30 KM/Hr
12	Turn	Medium	10.98686, 78.33767	30 KM/Hr
13	Turn	Medium	10.97264, 78.34383	30 KM/Hr
14	Turn	Medium	10.97205, 78.36594	30 KM/Hr
15	Turn	Medium	10.97076, 78.40086	30 KM/Hr
16	Turn	High	10.96016, 78.42212	15 KM/Hr
17	Turn	Medium	10.96079, 78.42266	30 KM/Hr
18	Turn	Medium	10.96084, 78.42307	30 KM/Hr
19	Blind Spot	Blind Spot	10.96050, 78.42371	10 KM/Hr
20	Turn	High	10.93502, 78.41317	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium
15	hospital	Aruthra Hospital	11.2264666, 78.1631524	30 km/h	Medium
16	clinic	Sri Ram Nursing Home, Namakkal	11.2259986, 78.1631944	30 km/h	Medium
17	clinic	Thangavel Skin Care	11.2253765, 78.1644957	30 km/h	Medium
18	hospital	Rudhra Hospital	11.226569, 78.163216	30 km/h	Medium
19	hospital	Rajendran Children's Hospital	11.2251887, 78.1649142	30 km/h	Medium
20	clinic	Lung Care Centre	11.2276789, 78.1638586	30 km/h	Medium
21	hospital	Vignesh Hospital	11.224818, 78.1621315	30 km/h	Medium
22	hospital	Venkat Hospital	11.2253645, 78.1646159	30 km/h	Medium
23	hospital	Shree Meenakshi Dental Hospital	11.225269, 78.164275	30 km/h	Medium
24	hospital	Annai Eye Hospital	11.2241645, 78.1641366	30 km/h	Medium
28	clinic	Kumaran Poly Clinic	11.2233138, 78.1639212	30 km/h	Medium
29	clinic	Chandra Test Tube Baby Centre	11.2241806, 78.1645931	30 km/h	Medium
30	clinic	Vishnu Ayurveda Clinic	11.224262, 78.164173	30 km/h	Medium
31	clinic	Dhanvanthiri Physiotheraphy	11.2245256, 78.1638528	30 km/h	Medium
32	hospital	Ilamurugan Hospital	11.2248399, 78.166586	30 km/h	Medium
33	hospital	Aburvaa Ayurveda Hospital	11.2228807, 78.1640453	30 km/h	Medium
34	hospital	Aravinth Hospital	11.2233336, 78.1672805	30 km/h	Medium
35	hospital	Sri Nidhi Hospital, Namakkal	11.2255711, 78.1687291	30 km/h	Medium
36	clinic	Athithya Nursing Home	11.2237726, 78.1686447	30 km/h	Medium
37	hospital	Kathiravan Hospital	11.226651, 78.167351	30 km/h	Medium
38	hospital	Sri Ramana Hospital	11.223503, 78.1685433	30 km/h	Medium
39	hospital	Vanitha Child Care & Skin Hospital	11.2243378, 78.16811	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
40	hospital	Sri Dhanvanthries Hospital	11.2241181, 78.1678614	30 km/h	Medium
41	hospital	Nirmala Hospital	11.2179361, 78.1665001	30 km/h	Medium
42	hospital	Surya Hospital, Namakkal	11.2178506, 78.1660375	30 km/h	Medium
43	hospital	Shanthi Hospitals, Namakkal	11.2190661, 78.1685735	30 km/h	Medium
44	hospital	Ganga Hospital	11.21742, 78.170272	30 km/h	Medium
45	hospital	Government Hospital	11.2168924, 78.169068	30 km/h	Medium
47	police	Police Station	11.2185353, 78.1695347	30 km/h	Medium
48	police	Women Police Station	11.2183133, 78.1692677	30 km/h	Medium
49	hospital	SMK Hospital	11.2172759, 78.1706866	30 km/h	Medium
50	hospital	Manoharan Hospital	11.218016, 78.171629	30 km/h	Medium
51	hospital	Saravana Hospital	11.216652, 78.171298	30 km/h	Medium
52	hospital	Latha Eye Hospital	11.2185689, 78.1719035	30 km/h	Medium
53	hospital	Vasan Eye Care Hospital, Namakkal	11.216301, 78.1717694	30 km/h	Medium
54	hospital	Aishwarya Hospital	11.2164042, 78.1715972	30 km/h	Medium
55	hospital	Poornima Hospital	11.215533, 78.16911	30 km/h	Medium
56	hospital	Nagarajan Hospital	11.2171673, 78.1718618	30 km/h	Medium
57	clinic	Ramesh ENT & Skin Care	11.217133, 78.171071	30 km/h	Medium
58	clinic	ENT Clinic	11.215786, 78.170545	30 km/h	Medium
59	clinic	Hari Chest Clinic	11.2162053, 78.1710093	30 km/h	Medium
60	hospital	Dr. Kandasamy Hospital	11.216051, 78.172052	30 km/h	Medium
61	clinic	Dr. Shanmugam ENT Clinic	11.218762, 78.171206	30 km/h	Medium
62	clinic	Children's Clinic	11.217004, 78.171504	30 km/h	Medium
63	clinic	Krishiv's Dental Care & Orthodontic Centre	11.2184886, 78.1713346	30 km/h	Medium
66	hospital	Sri Ragavendra Hospital	11.2151734, 78.1703431	30 km/h	Medium
67	hospital	Thangam Hospital,Namakkal	11.21472, 78.171857	30 km/h	Medium
68	clinic	Primecare Diabetic Clinic	11.213992, 78.1737091	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
72	hospital	Dr. A. Nagarajan Children's Hospital	11.212765, 78.1747068	30 km/h	Medium
73	hospital	M.M. Hospital	11.2103089, 78.1776545	30 km/h	Medium
74	clinic	Ram Clinic	11.2090922, 78.1790782	30 km/h	Medium
75	hospital	Imayam Speciality Hospital	11.2078899, 78.1797861	30 km/h	Medium
76	clinic	Prasith Clinic	11.208902, 78.180957	30 km/h	Medium
77	clinic	Aburvaa Multi Special Herbal Clinic	11.207568, 78.183483	30 km/h	Medium
78	hospital	Sugam Children's Hospital	11.2060083, 78.1854033	30 km/h	Medium
80	hospital	Government Hospital, Valayapatty	11.1212591, 78.2315962	30 km/h	Medium
83	hospital	Saravana Hospitals	10.9944313, 78.3273838	30 km/h	Medium
84	hospital	Dr. Siva Hospital	10.9877652, 78.3359228	30 km/h	Medium
85	hospital	Dr. Alva Hospital	10.987225, 78.336154	30 km/h	Medium
86	hospital	Ravikumar Hospital	10.988317, 78.336248	30 km/h	Medium
87	police	Thottiyam Police station	10.9874415, 78.3363001	30 km/h	Medium
88	hospital	Government Hospital, Thottiyam	10.98388, 78.338648	30 km/h	Medium
89	hospital	Government Hospital	10.9428055, 78.4146362	30 km/h	Medium
90	hospital	T M R Hospital	10.9429825, 78.4158242	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
12	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
13	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
14	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium
25	school	Government Girls School	11.2238954, 78.1662583	30 km/h	Medium
26	school	Government Elementary School	11.224555, 78.163967	30 km/h	Medium
27	school	Government Secondary School	11.2244426, 78.1647533	30 km/h	Medium
46	school	Government Girls Higher Secondary School	11.2179186, 78.1672383	30 km/h	Medium
64	school	Namakkal Government Boys Higher Secondary School	11.2150472, 78.1669273	30 km/h	Medium
65	college	Veterinary College and Research Institute Hospital	11.2144944, 78.1722653	30 km/h	Medium
69	college	Namakkal Kavignar Ramalingam Government Arts College for Women	11.2125137, 78.1724654	30 km/h	Medium
70	school	Kandasamy Kandar Matriculation and Higher Secondary School	11.2137988, 78.1698234	30 km/h	Medium
71	school	Selvam Matriculation Higher Secondary School	11.2124286, 78.1770824	30 km/h	Medium
79	college	King college	11.1692622, 78.2037212	30 km/h	Medium
81	school	Government Higher Secondary School	11.0234395, 78.2859835	30 km/h	Medium
82	college	Vetri vinayaga institution	11.013503, 78.3072053	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.94342, 78.41370



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44019, 77.87547



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39685, 77.88861



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38342, 77.89975



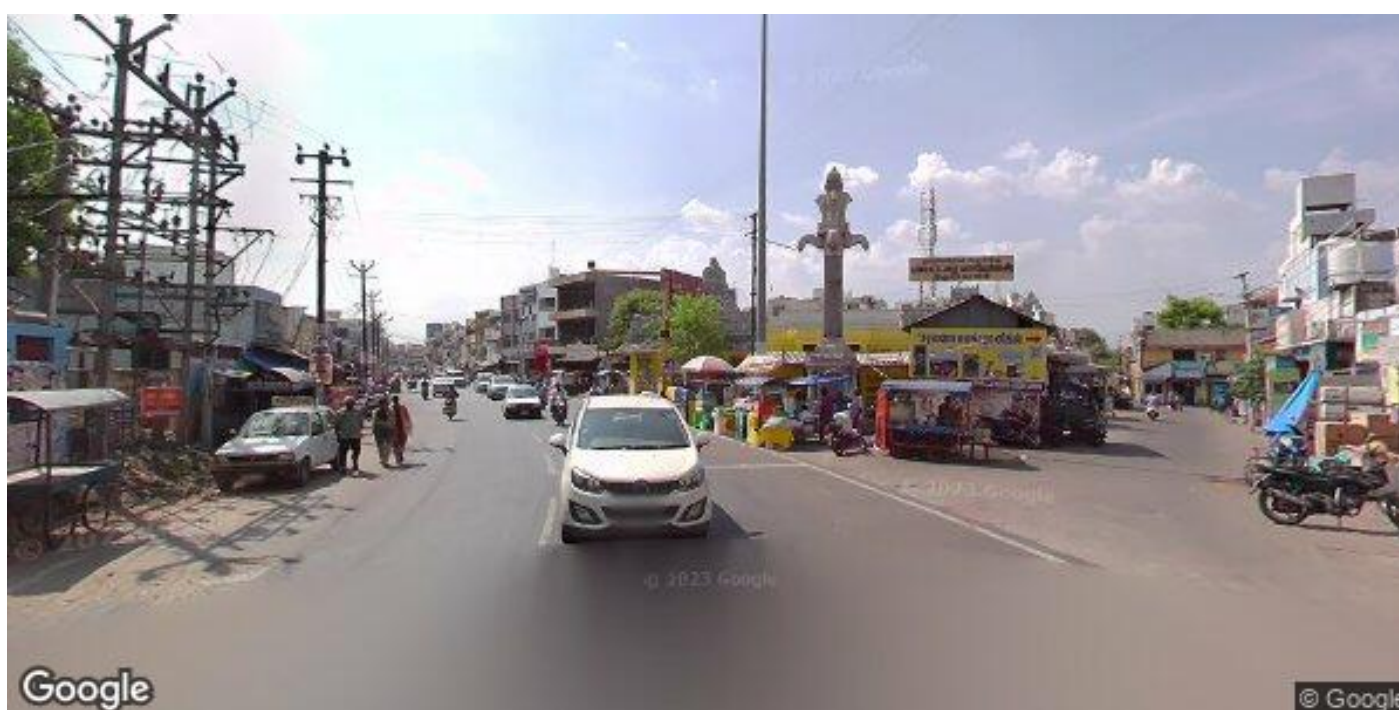
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38295, 77.90001



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38186, 77.90258



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.37841, 77.90140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.22411, 78.16637



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.12316, 78.23249



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.98686, 78.33767



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.97264, 78.34383



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.97205, 78.36594



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.97076, 78.40086



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.96079, 78.42266



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.96084, 78.42307



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 10.96050, 78.42371



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.93502, 78.41317

